Prof. Crocco, an Italian who migrated to the US, rejected me outright saying no more students. But I would not give up. I had a trump card up my sleeves: I had studied in great detail Crocco's stability theorem. It was my little secret that I had written to him while at TERLS seeking clarifications on his theorem. My mails never got a reply. When I kept pleading, he asked me what I expected: 'Have you read the Crocco theorem?' I was ready with a 'yes', but he would not be satisfied till I explained the theorem and answered his questions. I passed this exam and Prof. Crocco agreed to take me in with a condition—I had to do projects on campus and go to him at his home only for clarifications. Thrilled, I nodded in agreement.

As for the daily interaction on campus, I had to report to Prof. David T Harrje, who was as genial as he was demanding. There was a NASA project the department was handling and I grabbed the opportunity to work on it. It was on the behaviour of shock waves in a combustion zone, and was crucial for stability analyses of rocket systems. For someone who had attempted trials, errors and fun at the fledgling Indian space labs, this was high grade stuff.

The project involved sending a shock wave—a burst of air through a perforation on a membrane—after firing a liquid engine, and reading the behaviour of the wave. It was tough analysing the wave behaviour. It was pure mathematics, based on the readings. That may sound simple, till we realised that the number of variables here were more than the number of mathematical equations available. So you club the variables and analyse for the bunch. Well, this was complicated stuff.

I kept repeating the experiment well beyond the specified number of times. While this impressed Prof. Harrje, the technician on duty was not amused with the extra work he had to do in assisting me. He was not my concern, so the shock wave lived a hundred lives on the Princeton campus, firing my curiosity, quest and imagination, every time it passed through the perforation before dying its gradual yet quick death. After the initial reluctance, the technicians started spending longer hours with me, and we soon were friends.